Econometrics I

Lecture 11

Asymptotics

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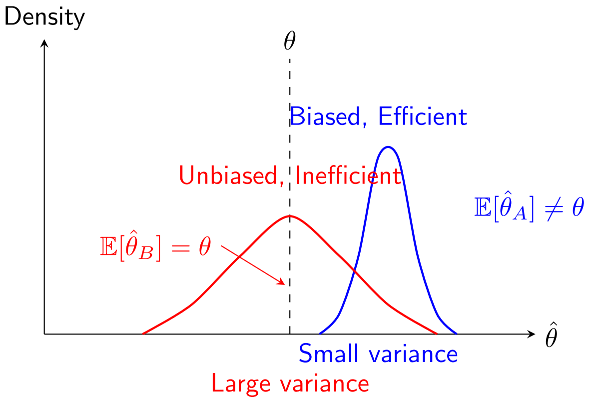
Introduction

What does asymptotic mean?

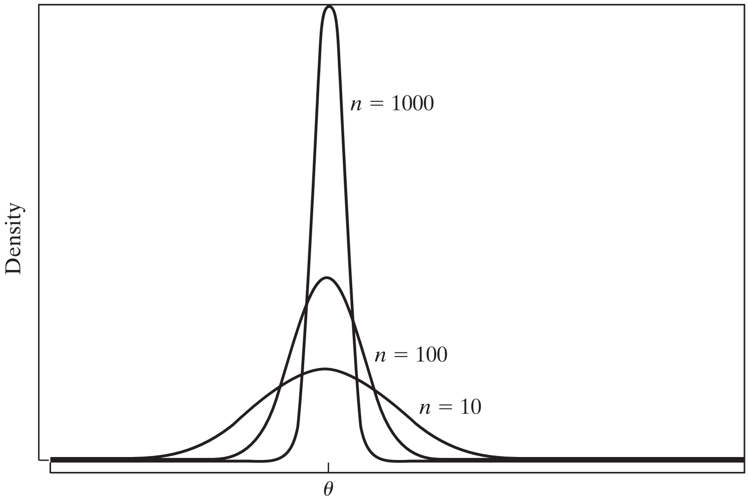
- Asymptotic = as $n \to \infty$ (i.e., when having infinitely many obs.)
- Serves as approximation for large sample scenario
- Key related concepts:
  - Consistency
  - Convergence in probability / in distribution
  - Asymptotic variance
  - Law of Large Numbers, Central Limit Theorem

Key concepts

Recall: Bias and variance



Consistency: the idea



Consistency: the idea

Consider following estimators:

1. $\hat{\theta}_1 = \theta + 0.001 \cdot n$
2. $\hat{\theta}_2 = \theta + 10,000/n$
3. $\hat{\theta}_3 = \theta + 0.001$

Which one is more efficient when $n = 10$?

Which one is biased?

Which one is consistent?

Consistency: the idea

- Many useful estimators are biased.
- **Consistency** = related concept: As $n \to \infty$, each $\hat{\beta}_j$ converges to the true population value β_j.
- Formally, this convergence is stochastic and called **convergence in probability**.
- We can use two equivalent notations for this:

$$\hat{\beta}_j \xrightarrow{p} \beta_j \qquad \Leftrightarrow \qquad \text{plim}\, \hat{\beta}_j = \beta_j$$

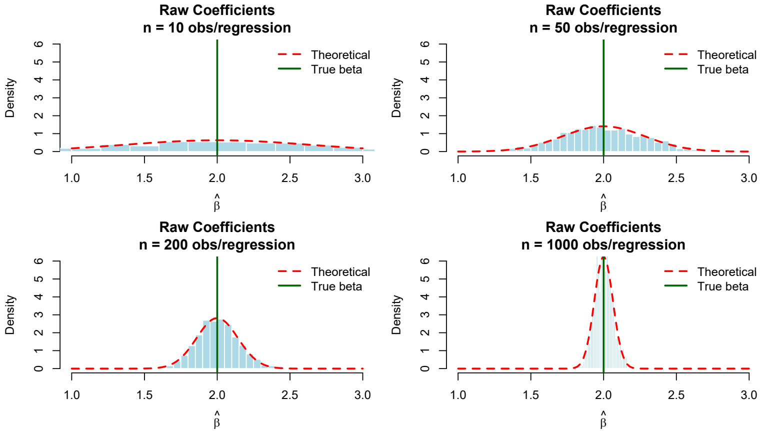
- The formal definition of this: For each (even tiny) $\varepsilon > 0$

$$\Pr\left(|\hat{\beta}_j - \beta_j| > \varepsilon\right) \to 0 \text{ as } n \to \infty$$

Consistency: the idea

- Consistency requires that
  - $\mathbb{E}(\hat{\beta}_j) \to \beta_j$ as $n \to \infty$
  - $\text{var}(\hat{\beta}_j) \to 0$ as $n \to \infty$
- Therefore
  - A biased estimator ($\mathbb{E}(\hat{\beta}_j) \neq \beta_j$) can be consistent if it is **asymptotically unbiased**, i.e. $\mathbb{E}(\hat{\beta}_j) \to \beta_j$ as $n \to \infty$ many highly relevant estimators have this feature.
  - An unbiased estimator can be inconsistent if $(\hat{\beta}_j) \nrightarrow 0$ as $n \to \infty$ some relevant estimators have this feature.

Consistency: the idea



Convergence in Probability and the LLN

Convergence in Probability:

$$X_n \xrightarrow{p} c \text{ if } \lim_{n\to\infty} P(|X_n - c| > \epsilon) = 0 \ \forall \epsilon > 0$$

Notation: $\text{plim}(X_n) = c$
Intuition: As n grows, X_n gets arbitrarily close to c with probability approaching 1.

Law of Large Numbers (LLN):

- If $X_1, \ldots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu$, then:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu$$

- Sample moments converge to population moments

Asymptotic variance

How **fast** (as $n \uparrow$) does the OLS variance converge to zero?

Recall, when $k = 2$ (univariate regression):

$$\text{var}(\hat{\beta}|x) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2}{n\,\text{var}(x)}$$

Note that variance falls at rate n

Hence, by variance-rule $\text{var}(a + b \cdot Y) = b^2\,\text{var}(Y)$:

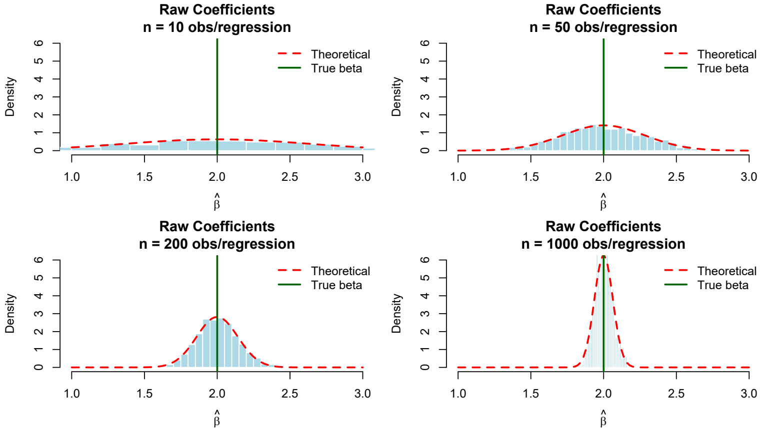
$$\text{var}(\hat{\beta} - \beta|x) = \frac{\sigma^2}{n\,\text{var}(x)}$$

$$\text{var}[\sqrt{n}(\hat{\beta} - \beta)|x] = (\sqrt{n})^2 \frac{\sigma^2}{n\,\text{var}(x)}$$

$$= \frac{\sigma^2}{\text{var}(x)} \qquad \textbf{root-n consistency!}$$

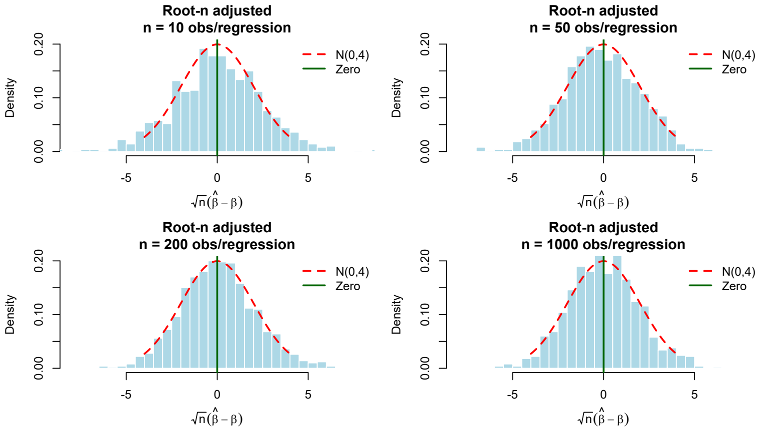
Asymptotic variance

Recall that variances of $\hat{\beta}$ get smaller with higher n:



Asymptotic variance

Now not $\hat{\beta}$ but $\sqrt{n}(\hat{\beta} - \beta)$:



Asymptotic normality

Convergence in Distribution and the CLT

Convergence in Distribution:

- $X_n \xrightarrow{d} X$ if CDF of X_n converges to CDF of X
- The *shape* of distribution stabilizes

Central Limit Theorem (CLT):

- If $X_1, \ldots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu$, $\text{var}(X_i) = \sigma^2$, then:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

- Equivalently: $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$
- Standardized sample mean becomes normal regardless of population distribution

What changes when $n \to \infty$?

Convergence rules (from Greene's textbook)

Rules for studying convergence:

THEOREM D.12 Slutsky Theorem
For a continuous function $g(x_n)$ that is not a function of n,

$$\text{plim}\, g(x_n) = g(\text{plim}\, x_n). \quad \textbf{(D-6)}$$

See Greene, App. D.

Convergence rules (from Greene's textbook)

THEOREM D.14 Rules for Probability Limits

If x_n and y_n are random variables with $\text{plim}\, x_n = c$ *and* $\text{plim}\, y_n = d$, *then*

$$\text{plim}(x_n + y_n) = c + d, \qquad \textbf{(sum rule)} \qquad \textbf{(D-7)}$$

$$\text{plim}\ x_n y_n = cd, \qquad \textbf{(product rule)} \qquad \textbf{(D-8)}$$

$$\text{plim}\ x_n / y_n = c/d \quad \text{if } d \neq 0. \qquad \textbf{(ratio rule)} \qquad \textbf{(D-9)}$$

If $\mathbf{W}_n$ *is a matrix whose elements are random variables and if* $\text{plim}\ \mathbf{W}_n = \mathbf{\Omega}$, *then*

$$\text{plim}\ \mathbf{W}_n^{-1} = \mathbf{\Omega}^{-1}. \qquad \textbf{(matrix inverse rule)} \qquad \textbf{(D-10)}$$

If $\mathbf{X}_n$ *and* $\mathbf{Y}_n$ *are random matrices with* $\text{plim}\ \mathbf{X}_n = \mathbf{A}$ *and* $\text{plim}\ \mathbf{Y}_n = \mathbf{B}$, *then*

$$\text{plim}\ \mathbf{X}_n \mathbf{Y}_n = \mathbf{AB}. \qquad \textbf{(matrix product rule)} \qquad \textbf{(D-11)}$$

Convergence rules (from Greene's textbook)

THEOREM D.16 Rules for Limiting Distributions

1. *If* $x_n \xrightarrow{d} x$ *and* plim $y_n = c$, *then*

$$x_n y_n \xrightarrow{d} cx, \tag{D-13}$$

which means that the limiting distribution of $x_n y_n$ *is the distribution of* cx. *Also,*

$$x_n + y_n \xrightarrow{d} x + c, \tag{D-14}$$

$$x_n / y_n \xrightarrow{d} x/c, \quad \text{if } c \neq 0. \tag{D-15}$$

2. *If* $x_n \xrightarrow{d} x$ *and* $g(x_n)$ *is a continuous function, then*

$$g(x_n) \xrightarrow{d} g(x). \tag{D-16}$$

This result is analogous to the Slutsky theorem for probability limits. For an example, consider the t_n *random variable discussed earlier. The exact distribution of* t_n^2 *is* $F[1, n]$. *But as* $n \longrightarrow \infty$, t_n *converges to a standard normal variable. According to this result, the limiting distribution of* t_n^2 *will be that of the square of a standard normal, which is chi-squared with one*

Implications

Let us first distinguish between **sample and population** quantities:

- **Sample covariance:** $\widehat{\text{cov}}(x,y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$
- **Population covariance:** $\text{cov}(x,y) = E[(x - E[x])(y - E[y])]$
- By LLN: $\widehat{\text{cov}}(x,y) \xrightarrow{p} \text{cov}(x,y)$
- Similarly: $\widehat{\text{var}}(x) \xrightarrow{p} \text{cov}(x)$

Hence, by convergence rules, in the simple model $y_i = \beta_1 + \beta_2 x_i + u_i$

$$\hat{\beta}_2 = \frac{\widehat{\text{cov}}(x,y)}{\widehat{\text{var}}(x)} \underset{A2}{=} \beta_2 + \frac{\widehat{\text{cov}}(x,u)}{\widehat{\text{var}}(x)}$$

we can take probability limits:

$$\text{plim}(\hat{\beta}_2) = \beta_2 + \frac{\text{plim }\widehat{\text{cov}}(x,u)}{\text{plim }\widehat{\text{var}}(x)} = \beta_2 + \frac{\text{cov}(x,u)}{\text{var}(x)} \underset{\text{cov}(x,u)=0}{=} \beta_2$$

Key conditions: $\boxed{\text{cov}(x,u) = 0, \mathbb{E}(u) = 0}$ (not the stronger requirement $\mathbb{E}[u|X] = 0$)

TAKEAWAYS

- No longer need of X-conditioning in $E(u|X) = 0$
  Sufficient: $\mathbb{E}(u) = 0$ and $\mathbb{E}(X'u) = 0$
- No longer need of normal errors assumption.
  With $\mathbb{E}(\mathbf{x}'\mathbf{x}) = \mathbf{A}$,

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \overset{a}{\sim} (\mathbf{0}, \sigma^2\mathbf{A}^{-1})$$

- Normal distribution instead of t-stat.
- Yet all other issues remain (omitted variable biases, heteroskedasticity, simultaneity, etc.).